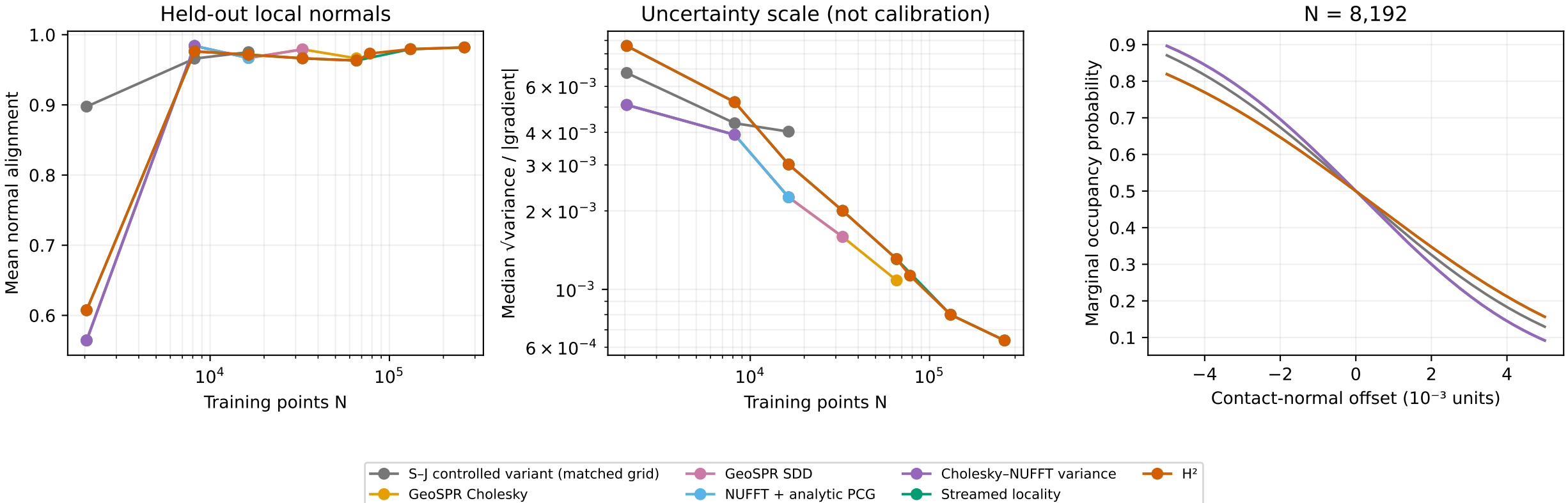
Held-out local normals
Mean normal alignment
1.0
0.9
0.8
0.7
0.6
10^4
10^5
Training points N
Uncertainty scale (not calibration)
Median √variance / |gradient|
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
10^{-3}
6×10^{-4}
10^4
10^5
Training points N
N = 8,192
Marginal occupancy probability
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
−4
−2
0
2
4
Contact-normal offset (10^{-3} units)
S–J controlled variant (matched grid)
GeoSPR Cholesky
GeoSPR SDD
NUFFT + analytic PCG
Cholesky–NUFFT variance
Streamed locality
H²